Preserving Visual Evidence in Compressed Multimodal Reasoning

Working Title

Preserving Visual Evidence in Compressed Multimodal Reasoning

Problem Statement

SketchThinker-R1 supervises reasoning style with a judge that receives neither the image nor the question, leaving the preservation of answer-critical visual evidence unverified. This motivates testing whether evidence-aware supervision can improve the accuracy–length tradeoff over style supervision.

Proposed Solution

We propose an image-conditioned sufficiency reward that favors concise traces retaining the visual premises and valid deductions needed to support an answer. Using one 7B model and verifiable diagram problems, we will annotate necessary premises and build human-checked image pairs that change one answer-critical fact; these annotations will train and validate a judge that sees the image, question, and trace, so a correct final answer alone does not count as sufficient. Sufficiency and style rewards will be compared on identical rewritten training data with matched compute, plus original-trace and length-preserving rewrite controls, reporting answer accuracy, premise retention, and paired-image correctness across token budgets and difficulty groups. This tests whether evidence preservation improves compression beyond effects of rewriting or training resources.

Key Research Questions

- Does text-only compression delete a larger proportion of annotated necessary visual premises than necessary nonvisual reasoning steps?
- At matched average reasoning-token budgets, does sufficiency supervision improve answer accuracy and necessary-premise retention over style supervision, particularly on harder items?
- Does the advantage persist with identical rewritten training data and matched training compute, and how do original-trace and length-preserving rewrite controls affect the comparison?

Expected Outcome

The study will deliver an evaluated evidence-preserving compression method and establish whether its accuracy–length tradeoff improves on style supervision under controlled comparisons.